

A Ternary $(4k - 1(2))/3$ Collatz-Type Map

Exact 3-Adic Symbolic Dynamics, a Density-One First-Descent Theorem,
Algebraic Cycle Analysis, and Exhaustive Computational Verification up to 10^{11}

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Abstract

We study the accelerated ternary Collatz-type map

$$\mathcal{S} = \{n \in \mathbb{Z}_{>0} : 3 \nmid n\}, \quad T(n) = \frac{4n - r(n)}{3\nu_3(4n - r(n))}, \quad r(n) = n \bmod 3 \in \{1, 2\}.$$

The map has two known positive fixed points, 1 and 2, and the known positive four-cycle

$$22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22.$$

The first part of the paper develops an exact 3-adic model rather than a heuristic one. On the full-measure forward-invariant domain $U^* \subset \mathbb{Z}_3^\times$, the inverse branches

$$\phi_{r,k}(y) = \frac{r + 3^k y}{4}, \quad r \in \{1, 2\}, \quad k \geq 1,$$

give an exact symbolic decomposition. With normalized Haar measure on $\mathbb{Z}_3^\times$, the successive valuations

$$a_i = \nu_3(4T^i(x) - r(T^i(x)))$$

are independent and identically distributed with

$$\mathbb{P}(a_i = k) = \frac{2}{3^k}, \quad k \geq 1.$$

Thus $\mathbb{E}[a_i] = 3/2$, $\text{Var}(a_i) = 3/4$, and the typical 3-adic valuation rate lies above the deterministic descent threshold $\alpha = \log_3 4$.

For an ordinary positive integer orbit $x_0 = n$, define $A_j = \sum_{i=0}^{j-1} a_i$. We prove the exact identity

$$T^j(n) = \frac{4^j n - C_j}{3^{A_j}}, \quad C_j = \sum_{i=0}^{j-1} r_i 4^{j-1-i} 3^{A_i} > 0,$$

and therefore the sufficient descent criterion

$$A_j \geq j \log_3 4 \implies T^j(n) < n.$$

Below this barrier we obtain the exact criterion

$$T^j(n) < n \iff C_j > (4^j - 3^{A_j})n.$$

A hypothetical orbit that never falls below its initial value must satisfy substantially stronger restrictions, including

$$\sum_{j \geq 0} \frac{3^{A_j}}{4^j} < \infty, \quad j \log_3 4 - A_j \rightarrow +\infty.$$

The 3-adic symbolic theory is then transferred rigorously to ordinary positive integers at every fixed depth. If

$$F_m = \{n \in \mathcal{S} : T^t(n) \geq n \text{ for } 1 \leq t \leq m\},$$

and $\mathcal{S}_m$ is the corresponding 3-adic valuation-barrier survivor set, then

$$\boxed{d_{\mathcal{S}}(F_m) = \mu(\mathcal{S}_m)}$$

for every fixed m . An exact-rational dynamic program computes these quantities. In particular,

$$\mu(\mathcal{S}_{50}) = \frac{301646097509840481608007680}{127173474825648610542883299603} \approx 0.002371926204921184320282,$$

so the relative natural density of admissible positive integers that achieve a first descent within 50 reduced steps is approximately 99.7628073795%. A Chernoff bound gives

$$\mu(\mathcal{S}_m) \leq \rho^m, \quad \rho \approx 0.952392652605812657 < 1,$$

and yields the main density theorem

$$\boxed{d_{\mathcal{S}}\{n \in \mathcal{S} : \exists j \geq 1, T^j(n) < n\} = 1.}$$

This is a density-one first-descent theorem; it is not a theorem of universal first descent or global convergence.

Finally, the archived exhaustive computation through 10^{11} contains exactly 66,666,666,667 admissible starts. All are classified into the three known positive terminal cycles, with zero reported unknown cycles and zero arithmetic overflows. The basin counts are 6,602,925,203 for $\{1\}$, 43,211,124,603 for $\{2\}$, and 16,852,616,861 for the four-cycle. Moreover, every admissible n with $50 < n \leq 10^{11}$ has a strict first descent. The largest observed first-descent time is 355, attained at $n = 76,089,719,024$. The complete 10^{11} archive was independently re-aggregated and structurally audited; a hardened V5.2 scanner was freshly rerun on validation ranges and on a 100-million-integer production interval containing the 355-step record. The full 10^{11} domain was not freshly recomputed from zero with V5.2 during that audit.

The global conjecture that every positive admissible integer eventually enters one of the three known positive terminal cycles remains open.

Keywords: Collatz-type map; ternary dynamics; 3-adic valuation; Haar measure; symbolic dynamics; first descent; natural density; dynamic programming; Diophantine cycles; exhaustive computation.

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1 Introduction

The classical $3x + 1$ problem is a model example of an integer dynamical system whose rule is elementary but whose global behavior is difficult to control. The main source of difficulty is the interaction between an expanding affine operation and a residue-dependent division. Generalized Collatz maps retain this interaction while changing the modulus, affine coefficients, or acceleration convention; see, for example, Lagarias [1] and Carnielli [2] for background on the classical problem and related generalizations.

The map studied here is ternary rather than binary. Its state space consists of positive integers not divisible by 3. At each reduced step we multiply by 4, subtract the current nonzero residue modulo 3, and then remove *all* factors of 3. Thus one reduced iteration combines an affine expansion with a variable amount of 3-adic compression.

An earlier version of this study established the basic algebra of the map and reported exhaustive finite computations through 10^9 and later through 10^{11} [3]. The present version substantially strengthens the theoretical part. The geometric law for the removed 3-adic valuation is no longer treated merely as a heuristic: on the natural 3-adic unit space it follows exactly from the inverse branches and normalized Haar measure. More importantly, the resulting finite-horizon barrier probabilities can be transferred rigorously to relative natural densities of ordinary positive integers. This produces an exact finite-horizon density theorem and, after a Chernoff estimate, a density-one first-descent theorem.

The paper deliberately separates four logically different levels of information.

1. **Exact 3-adic theorems.** These concern Haar measure on the 3-adic unit space and establish the exact symbolic law of residues and valuations.
2. **Exact deterministic orbit identities.** These hold for every ordinary positive integer orbit and give the descent barrier and necessary conditions for a hypothetical no-descent orbit.
3. **Natural-density theorems.** These transfer fixed-depth 3-adic cylinder information to asymptotic relative natural density among ordinary positive integers.
4. **Finite exhaustive computation.** This verifies the behavior of every admissible start through 10^{11} , but makes no assertion about untested larger integers.

Keeping these levels separate is essential. A Haar-almost-everywhere statement on $\mathbb{Z}_3^\times$ does not by itself imply a statement about every positive integer. A density-one statement does not imply that the exceptional set is empty. A finite exhaustive computation, no matter how large, does not prove an all-integers conjecture.

The principal rigorous results of this paper are:

- an exact inverse-branch decomposition of $\mathbb{Z}_3^\times$;
- the iid valuation law $\mathbb{P}(a_i = k) = 2/3^k$;
- the exact multi-step identity for $T^j(n)$;
- the sufficient descent barrier $A_j \geq j \log_3 4$;
- exact necessary conditions for any hypothetical orbit that never descends below its start;
- the fixed-depth density identity $d_S(F_m) = \mu(\mathcal{S}_m)$;
- the density-one first-descent theorem;
- exact-rational finite-depth values through depth 500;
- exhaustive finite verification through 10^{11} and an exhaustive first-descent verification for all admissible $50 < n \leq 10^{11}$.

The global convergence conjecture remains open. In particular, this paper does not prove universal first descent, density-one convergence to the known cycles, absence of all other positive cycles at arbitrary size, or absence of unbounded positive trajectories.

2 The map, notation, and elementary examples

2.1 Positive state space

Define

$$\mathcal{S} = \{x \in \mathbb{Z}_{>0} : 3 \nmid x\}. \quad (1)$$

Every $x \in \mathcal{S}$ has a unique residue

$$r(x) = x \bmod 3 \in \{1, 2\}. \quad (2)$$

Define the affine numerator

$$M(x) = 4x - r(x) = \begin{cases} 4x - 1, & x \equiv 1 \pmod{3}, \\ 4x - 2, & x \equiv 2 \pmod{3}. \end{cases} \quad (3)$$

Since $4 \equiv 1 \pmod{3}$, one always has

$$M(x) \equiv x - r(x) \equiv 0 \pmod{3}.$$

Therefore $M(x)$ contains at least one factor of 3.

For a nonzero integer m , write

$$\nu_3(m) = \max\{a \geq 0 : 3^a \mid m\}.$$

The accelerated map is

$$T(x) = \frac{4x - r(x)}{3^{\nu_3(4x - r(x))}}. \quad (4)$$

Because all factors of 3 are removed, $T(x) \in \mathcal{S}$. Thus $T : \mathcal{S} \rightarrow \mathcal{S}$.

2.2 Meaning of the title notation

The compact title notation $(4k - 1(2))/3$ is shorthand for the two residue branches: subtract 1 for a state congruent to 1 $\pmod{3}$ and subtract 2 for a state congruent to 2 $\pmod{3}$. The precise map is always (4). In particular, the map divides by the complete power $3^{\nu_3(M(x))}$, not merely by one factor of 3.

2.3 Residue-form branches

If $x = 3q + 1$, then

$$M(x) = 4(3q + 1) - 1 = 12q + 3 = 3(4q + 1),$$

and hence

$$T(3q + 1) = \frac{4q + 1}{3^{\nu_3(4q+1)}}. \quad (5)$$

If $x = 3q + 2$, then

$$M(x) = 4(3q + 2) - 2 = 12q + 6 = 3(4q + 2),$$

and

$$T(3q + 2) = \frac{4q + 2}{3^{\nu_3(4q+2)}}. \quad (6)$$

These formulas make the acceleration transparent: one factor of 3 is forced by the residue choice, while additional factors depend on the smaller branch quantity.

2.4 Small numerical examples

Example 2.1 (The fixed point 1). For $x = 1$,

$$M(1) = 4 - 1 = 3, \quad \nu_3(M(1)) = 1,$$

so

$$T(1) = 3/3 = 1.$$

Thus $\{1\}$ is a fixed point.

Example 2.2 (The fixed point 2). For $x = 2$,

$$M(2) = 8 - 2 = 6, \quad \nu_3(M(2)) = 1,$$

and

$$T(2) = 6/3 = 2.$$

Thus $\{2\}$ is a second fixed point.

Example 2.3 (A short trajectory). Starting at 4,

$$4 \mapsto \frac{16 - 1}{3} = 5,$$

while

$$5 \mapsto \frac{20 - 2}{3^2} = 2.$$

Hence

$$4 \rightarrow 5 \rightarrow 2.$$

Example 2.4 (A large one-step valuation). For $x = 7$,

$$M(7) = 28 - 1 = 27 = 3^3,$$

so

$$T(7) = 1.$$

A valuation $a = 3$ overwhelms the factor 4 and causes a sharp decrease.

Example 2.5 (The positive four-cycle). The orbit

$$22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22$$

is verified step by step:

$$4(22) - 1 = 87 = 3 \cdot 29,$$

$$4(29) - 2 = 114 = 3 \cdot 38,$$

$$4(38) - 2 = 150 = 3 \cdot 50,$$

and

$$4(50) - 2 = 198 = 3^2 \cdot 22.$$

The valuation word is therefore $(1, 1, 1, 2)$ and its total valuation is 5.

3 The 3-adic unit space

The exact probability law used later is most naturally proved on the 3-adic unit space.

3.1 Basic 3-adic setting

Let

$$U = \mathbb{Z}_3^\times = \{x \in \mathbb{Z}_3 : 3 \nmid x\}.$$

The same residue map $r(x) \in \{1, 2\}$ is defined by reduction modulo 3. Whenever $4x - r(x) \neq 0$, put

$$a(x) = \nu_3(4x - r(x)) \geq 1, \quad T(x) = \frac{4x - r(x)}{3^{a(x)}}.$$

The quotient is again a 3-adic unit.

Let m be additive Haar probability measure on $\mathbb{Z}_3$, so $m(\mathbb{Z}_3) = 1$. Since

$$U = (1 + 3\mathbb{Z}_3) \sqcup (2 + 3\mathbb{Z}_3),$$

we have $m(U) = 2/3$. We use normalized Haar probability measure on U :

$$\mu(B) = \frac{m(B)}{m(U)} = \frac{3}{2}m(B). \tag{7}$$

3.2 Exceptional zero-numerator points

The numerator vanishes only when

$$4x - r(x) = 0.$$

For $r = 1$ this gives $x = 1/4$; for $r = 2$ it gives $x = 1/2$. Both are 3-adic units. Put

$$Z_0 = \left\{ \frac{1}{4}, \frac{1}{2} \right\}.$$

Let

$$Z = \bigcup_{j \geq 0} T^{-j}(Z_0). \quad (8)$$

Each point has at most countably many one-step preimages, so every finite backward level of Z_0 is countable, and therefore Z is countable. Haar measure is nonatomic, hence

$$\mu(Z) = 0.$$

Define the full-measure domain

$$U^* = U \setminus Z. \quad (9)$$

Then $\mu(U^*) = 1$, every forward iterate is defined on U^* , and $T(U^*) \subseteq U^*$.

This removal is a technical 3-adic issue. No positive integer in $\mathcal{S}$ is removed, because for a positive integer x the numerator $4x - r(x)$ is strictly positive.

4 Exact inverse branches and Haar scaling

For $r \in \{1, 2\}$ and $k \geq 1$, define

$$\phi_{r,k}(y) = \frac{r + 3^k y}{4}. \quad (10)$$

Lemma 4.1 (Exact inverse branches). *For every $r \in \{1, 2\}$, $k \geq 1$, and $y \in U$,*

$$\phi_{r,k}(y) \in U, \quad r(\phi_{r,k}(y)) = r, \quad a(\phi_{r,k}(y)) = k, \quad T(\phi_{r,k}(y)) = y.$$

Proof. Because 4 is a unit in $\mathbb{Z}_3$, the quantity $\phi_{r,k}(y)$ belongs to $\mathbb{Z}_3$. Modulo 3,

$$4\phi_{r,k}(y) = r + 3^k y \equiv r \pmod{3}.$$

Since $4 \equiv 1 \pmod{3}$, we obtain

$$\phi_{r,k}(y) \equiv r \pmod{3},$$

so $\phi_{r,k}(y)$ is a unit with residue label r . Moreover,

$$4\phi_{r,k}(y) - r = 3^k y.$$

Because y is a unit, $\nu_3(y) = 0$, and therefore the valuation of the numerator is exactly k . Dividing by 3^k gives $T(\phi_{r,k}(y)) = y$. $\square$

Lemma 4.2 (Branch partition). *Apart from the two zero-numerator points,*

$$U \setminus Z_0 = \bigsqcup_{r \in \{1, 2\}} \bigsqcup_{k \geq 1} \phi_{r,k}(U). \quad (11)$$

Proof. Take $x \in U \setminus Z_0$. Its residue $r = r(x)$ is unique, and the nonzero numerator $4x - r$ has a unique finite valuation

$$k = \nu_3(4x - r) \geq 1.$$

Set

$$y = \frac{4x - r}{3^k} \in U.$$

Then $x = \phi_{r,k}(y)$. This proves completeness. Uniqueness of the residue and of the exact valuation proves disjointness. $\square$

Lemma 4.3 (Exact Haar scaling). *For every measurable $B \subseteq U$,*

$$\boxed{\mu(\phi_{r,k}(B)) = 3^{-k} \mu(B).} \quad (12)$$

Proof. The affine map can be written

$$\phi_{r,k}(y) = \frac{r}{4} + \frac{3^k}{4}y.$$

Translation preserves additive Haar measure. Multiplication by $3^k/4$ scales it by its 3-adic absolute value

$$\left| \frac{3^k}{4} \right|_3 = 3^{-k}.$$

Hence

$$m(\phi_{r,k}(B)) = 3^{-k} m(B).$$

Normalizing by $m(U) = 2/3$ gives (12). $\square$

5 Exact symbolic law and iid valuations

Theorem 5.1 (One-step valuation law). *For $k \geq 1$,*

$$\boxed{\mu(a = k) = \frac{2}{3^k}.} \quad (13)$$

More precisely,

$$\boxed{\mu(r = r_0, a = k) = 3^{-k}} \quad (14)$$

for each $r_0 \in \{1, 2\}$.

Proof. For fixed (r, k) , the corresponding branch image has measure 3^{-k} by (12). The two residue branches are disjoint, so

$$\mu(a = k) = 2 \cdot 3^{-k}.$$

The normalization check is

$$\sum_{k=1}^{\infty} \frac{2}{3^k} = 1.$$

$\square$

Proposition 5.2 (Haar invariance). *For every measurable $B \subseteq U^*$,*

$$\boxed{\mu(T^{-1}B) = \mu(B).} \quad (15)$$

Proof. Using the disjoint inverse branches,

$$T^{-1}B = \bigsqcup_{r=1}^2 \bigsqcup_{k \geq 1} \phi_{r,k}(B).$$

Therefore

$$\mu(T^{-1}B) = \sum_{r=1}^2 \sum_{k \geq 1} 3^{-k} \mu(B) = 2 \left(\sum_{k \geq 1} 3^{-k} \right) \mu(B) = \mu(B).$$

□

For $i \geq 0$, define

$$r_i(x) = r(T^i x), \quad a_i(x) = a(T^i x).$$

Theorem 5.3 (Exact finite cylinders and iid symbolic dynamics). *Fix $m \geq 1$, residues $r_i \in \{1, 2\}$, and valuations $k_i \geq 1$ for $0 \leq i < m$. The corresponding symbol cylinder has measure*

$$\mu\{x : (r_i(x), a_i(x)) = (r_i, k_i), \ 0 \leq i < m\} = 3^{-\sum_{i=0}^{m-1} k_i}. \quad (16)$$

Consequently,

$$\mu(a_0 = k_0, \dots, a_{m-1} = k_{m-1}) = \prod_{i=0}^{m-1} \frac{2}{3^{k_i}}. \quad (17)$$

Thus $(a_i)_{i \geq 0}$ are iid with law (13); indeed the pairs (r_i, a_i) are iid with

$$\mathbb{P}(r_i = r, a_i = k) = 3^{-k}.$$

Proof. The full symbol cylinder is the iterated inverse image

$$\phi_{r_0, k_0} \circ \phi_{r_1, k_1} \circ \dots \circ \phi_{r_{m-1}, k_{m-1}}(U^*).$$

Repeated application of (12) gives

$$3^{-(k_0 + \dots + k_{m-1})}.$$

For a fixed valuation word $(k_0, \dots, k_{m-1})$, there are 2^m disjoint residue words. Summing their equal measures yields

$$2^m 3^{-\sum_i k_i} = \prod_i \frac{2}{3^{k_i}},$$

which is exactly the product law. □

5.1 Moments and the typical valuation rate

Using the standard geometric sums,

$$\mathbb{E}[a] = \sum_{k \geq 1} k \frac{2}{3^k} = \frac{3}{2},$$

$$\mathbb{E}[a^2] = \sum_{k \geq 1} k^2 \frac{2}{3^k} = 3,$$

and therefore

$$\text{Var}(a) = 3 - \left(\frac{3}{2}\right)^2 = \frac{3}{4}. \quad (18)$$

Let

$$A_j = \sum_{i=0}^{j-1} a_i.$$

By the Strong Law of Large Numbers,

$$\frac{A_j}{j} \longrightarrow \frac{3}{2} \quad \text{for Haar-almost every } x \in U^*. \quad (19)$$

The critical deterministic slope introduced below is

$$\alpha = \log_3 4 \approx 1.261859507142914874.$$

Since $3/2 > \alpha$, a Haar-typical symbolic path eventually lies above the deterministic valuation barrier.

Remark 5.4 (What this does and does not mean). There is no order relation on $\mathbb{Z}_3$ compatible with the ordinary inequality $<$ on positive integers, so (19) should not be described as “3-adic descent.” Moreover, the positive integers form a countable Haar-null subset of $\mathbb{Z}_3$. Therefore an almost-everywhere statement on U^* does not by itself imply anything pointwise about a particular positive integer. The transfer to ordinary integers is proved separately in Sections 10–13.

6 Exact deterministic iteration identity

Return now to an ordinary positive orbit

$$x_0 = n \in \mathcal{S}, \quad x_{i+1} = T(x_i).$$

Write

$$r_i = r(x_i) \in \{1, 2\}, \quad a_i = \nu_3(4x_i - r_i) \geq 1.$$

Then

$$3^{a_i} x_{i+1} = 4x_i - r_i. \quad (20)$$

Define

$$A_0 = 0, \quad A_j = \sum_{i=0}^{j-1} a_i. \quad (21)$$

Theorem 6.1 (Exact multi-step identity). *For every $j \geq 0$,*

$$\boxed{x_j = \frac{4^j n - C_j}{3^{A_j}}}, \quad (22)$$

where

$$C_0 = 0, \quad \boxed{C_{j+1} = 4C_j + r_j 3^{A_j}}, \quad (23)$$

and, equivalently,

$$\boxed{C_j = \sum_{i=0}^{j-1} r_i 4^{j-1-i} 3^{A_i}}. \quad (24)$$

In particular, $C_j > 0$ for every $j \geq 1$.

Proof. The case $j = 0$ is immediate. Assume

$$3^{A_j} x_j = 4^j n - C_j.$$

Multiplying (20) by 3^{A_j} gives

$$3^{A_{j+1}} x_{j+1} = 4(3^{A_j} x_j) - r_j 3^{A_j}.$$

Substitution of the induction hypothesis yields

$$3^{A_{j+1}} x_{j+1} = 4^{j+1} n - (4C_j + r_j 3^{A_j}).$$

This proves (22) and (23). Expanding the recurrence from $C_0 = 0$ gives (24); every term in that sum is positive. $\square$

Example 6.2 (Two-step identity for $n = 4$). The orbit is $4 \rightarrow 5 \rightarrow 2$. Here

$$(r_0, a_0) = (1, 1), \quad (r_1, a_1) = (2, 2),$$

so $A_2 = 3$. Also

$$C_2 = r_0 4 + r_1 3^{A_1} = 1 \cdot 4 + 2 \cdot 3 = 10.$$

Hence

$$T^2(4) = \frac{4^2 \cdot 4 - 10}{3^3} = \frac{54}{27} = 2,$$

as required.

7 The deterministic descent barrier

Set

$$\alpha = \log_3 4 = \frac{\log 4}{\log 3} \approx 1.261859507142914874. \quad (25)$$

Theorem 7.1 (Sufficient descent barrier). *If for some $j \geq 1$,*

$$3^{A_j} \geq 4^j, \quad (26)$$

equivalently

$$A_j \geq \alpha j, \quad (27)$$

then

$$\boxed{T^j(n) < n.} \quad (28)$$

Proof. By (22) and $C_j > 0$,

$$T^j(n) = \frac{4^j n - C_j}{3^{A_j}} < \frac{4^j n}{3^{A_j}}.$$

Under (26), the right-hand side is at most n . Therefore $T^j(n) < n$. $\square$

The equality $3^{A_j} = 4^j$ cannot occur for positive integer j, A_j by unique prime factorization, but even if it could, the positive correction C_j would still make the descent strict.

7.1 Exact criterion below the barrier

Suppose

$$3^{A_j} < 4^j.$$

Then (22) gives

$$T^j(n) < n \iff 4^j n - C_j < 3^{A_j} n.$$

Theorem 7.2 (Exact below-barrier descent criterion). *When $3^{A_j} < 4^j$,*

$$\boxed{T^j(n) < n \iff C_j > (4^j - 3^{A_j})n.} \quad (29)$$

Equivalently, with

$$R_j = \frac{C_j}{4^j - 3^{A_j}}, \quad (30)$$

one has

$$\boxed{T^j(n) < n \iff n < R_j.} \quad (31)$$

Thus crossing the valuation barrier is sufficient, not necessary. The positive correction term C_j can cause descent even when $A_j < \alpha j$.

7.2 Excess-valuation form

Because every $a_i \geq 1$, define

$$E_j = A_j - j = \sum_{i=0}^{j-1} (a_i - 1)$$

and

$$\beta = \alpha - 1 \approx 0.261859507142914874.$$

The sufficient barrier is equivalently

$$E_j \geq \beta j. \quad (32)$$

A path that never triggers the sufficient barrier must satisfy

$$E_j < \beta j$$

at every prefix.

8 Necessary structure of a hypothetical no-descent orbit

Assume, for the rest of this section only, the hypothesis

$$\boxed{x_j = T^j(n) \geq n \quad \text{for every } j \geq 1.} \quad (\text{H})$$

This is an *eternal no-descent hypothesis*. We derive necessary consequences but no contradiction.

Corollary 8.1 (Barrier avoidance). *Under (H),*

$$A_j < \alpha j \quad \text{for every } j \geq 1, \quad (33)$$

and

$$C_j \leq (4^j - 3^{A_j})n. \quad (34)$$

Proof. The first statement is the contrapositive of the sufficient descent theorem. The second follows from

$$4^j n - C_j = 3^{A_j} x_j \geq 3^{A_j} n.$$

□

Define

$$q_j = \frac{3^{A_j}}{4^j} = 3^{A_j - \alpha j}. \quad (35)$$

Under (H), $0 < q_j < 1$ for $j \geq 1$, and $q_0 = 1$.

Dividing

$$4^j n = C_j + 3^{A_j} x_j$$

by 4^j and using (24) gives the exact normalized identity

$$n = \frac{1}{4} \sum_{i=0}^{j-1} r_i q_i + q_j x_j. \quad (36)$$

Theorem 8.2 (Summability under eternal no descent). *Under (H),*

$$\sum_{i=0}^{\infty} q_i = \sum_{i=0}^{\infty} \frac{3^{A_i}}{4^i} < \infty. \quad (37)$$

Consequently,

$$q_j \rightarrow 0. \quad (38)$$

Proof. The last term in (36) is positive, so

$$\frac{1}{4} \sum_{i=0}^{j-1} r_i q_i < n.$$

Since $r_i \geq 1$,

$$\sum_{i=0}^{j-1} q_i < 4n.$$

The partial sums are increasing and uniformly bounded, proving convergence. Every term of a convergent positive series tends to zero. □

Define the deficit from the critical line

$$D_j = \alpha j - A_j. \quad (39)$$

Then

$$q_j = 3^{-D_j}.$$

Theorem 8.3 (Unbounded valuation deficit). *Under (H),*

$$D_j = \alpha j - A_j \longrightarrow +\infty. \quad (40)$$

Equivalently,

$$\beta j - \sum_{i=0}^{j-1} (a_i - 1) \longrightarrow +\infty. \quad (41)$$

Proof. By (38), $3^{-D_j} \rightarrow 0$, which is equivalent to $D_j \rightarrow +\infty$. $\square$

A simple frequency consequence is worth recording. Let

$$N_{\geq 2}(j) = \#\{0 \leq i < j : a_i \geq 2\}.$$

Because each $a_i \geq 2$ contributes at least one to $\sum(a_i - 1)$,

$$N_{\geq 2}(j) \leq \sum_{i < j} (a_i - 1) < \beta j.$$

Thus more than approximately

$$1 - \beta \approx 0.738140493$$

of the first j valuations must equal 1 at every prefix of a hypothetical no-descent orbit. This is a necessary condition, not a contradiction.

8.1 A decreasing normalized quantity

Define

$$y_j = q_j x_j = \frac{3^{A_j} x_j}{4^j}. \quad (42)$$

Then (36) becomes

$$n = \frac{1}{4} \sum_{i=0}^{j-1} r_i q_i + y_j. \quad (43)$$

Subtracting the identity at consecutive indices gives

$$\boxed{y_{j+1} = y_j - \frac{r_j}{4} q_j.} \quad (44)$$

Hence y_j is positive and strictly decreasing. Therefore

$$L = \lim_{j \rightarrow \infty} y_j \quad (45)$$

exists with $L \geq 0$. Passing to the limit in (43),

$$\boxed{n = \frac{1}{4} \sum_{i=0}^{\infty} r_i q_i + L.} \quad (46)$$

8.2 Product representation and the two limiting regimes

From the one-step relation,

$$q_{j+1} = q_j \frac{3^{a_j}}{4}.$$

Therefore

$$y_{j+1} = q_{j+1} x_{j+1} = q_j x_j \left(1 - \frac{r_j}{4x_j}\right).$$

Thus

$$\boxed{y_j = n \prod_{i=0}^{j-1} \left(1 - \frac{r_i}{4x_i}\right).} \quad (47)$$

Since the factors lie strictly between 0 and 1, standard infinite-product criteria give

$$L > 0 \iff \sum_{i=0}^{\infty} \frac{r_i}{x_i} < \infty \iff \sum_{i=0}^{\infty} \frac{1}{x_i} < \infty. \quad (48)$$

If $L > 0$, then $q_j \rightarrow 0$ and

$$x_j = \frac{y_j}{q_j} \sim \frac{L}{q_j} = L3^{D_j},$$

so

$$x_j \rightarrow \infty. \quad (49)$$

Thus the $L > 0$ regime would be an escaping no-descent orbit.

If $L = 0$, then (46) becomes the exact identity

$$n = \frac{1}{4} \sum_{i=0}^{\infty} r_i \frac{3^{A_i}}{4^i}. \quad (50)$$

Neither regime is presently contradictory. In particular, $q_j \rightarrow 0$ does *not* imply $y_j \rightarrow 0$, because x_j may grow at a compensating rate.

8.3 Threshold limit and congruence

Under (H),

$$R_j = \frac{\frac{1}{4} \sum_{i=0}^{j-1} r_i q_i}{1 - q_j},$$

so

$$R_j \rightarrow n - L. \quad (51)$$

The multi-step identity also gives the exact congruence

$$4^j n \equiv C_j \pmod{3^{A_j}}, \quad (52)$$

or, since 4^j is invertible modulo 3^{A_j} ,

$$n \equiv 4^{-j} C_j \pmod{3^{A_j}}.$$

These congruences encode finite orbit prefixes but do not by themselves provide an additional obstruction to an infinite orbit.

Remark 8.4 (Boundary of the deterministic argument). The preceding results do not prove that $L = 0$, do not derive a contradiction from either $L = 0$ or $L > 0$, do not exclude every infinite low-valuation word, and do not prove global convergence. They characterize what a counterexample would have to satisfy.

9 Algebraic equation for periodic orbits

Suppose a periodic orbit has length ℓ :

$$x_0, x_1, \dots, x_{\ell-1}, \quad x_{\ell} = x_0.$$

Let

$$r_i = r(x_i), \quad k_i = \nu_3(4x_i - r_i), \quad K = \sum_{i=0}^{\ell-1} k_i.$$

Applying the multi-step identity at $j = \ell$ gives

$$x_0 = \frac{4^\ell x_0 - C_\ell}{3^K},$$

and therefore

$$x_0 = \frac{\sum_{j=0}^{\ell-1} r_j 4^{\ell-1-j} 3^{\sum_{m=0}^{j-1} k_m}}{4^\ell - 3^K}. \quad (53)$$

The empty exponent sum is interpreted as 0.

This formula gives an exact Diophantine condition. A proposed residue/valuation word must make the right-hand side an integer, and the resulting orbit must reproduce the assumed residues and valuations.

9.1 Sign condition

The numerator in (53) is positive. Therefore a positive cycle requires

$$4^\ell - 3^K > 0 \iff \frac{K}{\ell} < \log_3 4 = \alpha. \quad (54)$$

A negative cycle, when the map is algebraically extended to negative states, requires the opposite sign. Thus rational approximations K/ℓ to $\log_3 4$ are naturally connected to small cycle denominators.

9.2 Reconstruction of the known positive cycles

For a fixed point with $k_0 = 1$, (53) gives

$$x_0 = \frac{r_0}{4 - 3} = r_0,$$

yielding $x_0 = 1$ and $x_0 = 2$.

For the four-cycle,

$$\begin{aligned} (r_0, r_1, r_2, r_3) &= (1, 2, 2, 2), \\ (k_0, k_1, k_2, k_3) &= (1, 1, 1, 2), \quad K = 5. \end{aligned}$$

The denominator is

$$4^4 - 3^5 = 256 - 243 = 13.$$

The numerator is

$$1 \cdot 4^3 + 2 \cdot 4^2 \cdot 3 + 2 \cdot 4 \cdot 3^2 + 2 \cdot 3^3 = 64 + 96 + 72 + 54 = 286.$$

Hence

$$x_0 = \frac{286}{13} = 22.$$

The small denominator reflects the near equality

$$4^4 = 256, \quad 3^5 = 243.$$

The near equality is arithmetically favorable but is not sufficient by itself; exact divisibility and branch consistency are still required.

10 Finite-horizon survivor sets and transfer to integers

The 3-adic iid law becomes useful for ordinary integers only after a separate density-transfer argument.

10.1 Relative natural density

For $A \subseteq \mathcal{S}$, define the relative natural density, when it exists, by

$$d_{\mathcal{S}}(A) = \lim_{N \rightarrow \infty} \frac{\#(A \cap [1, N])}{\#(\mathcal{S} \cap [1, N])}. \quad (55)$$

This normalizes density relative to the admissible integers rather than to all positive integers.

For $x \in U^*$, let

$$A_t(x) = \sum_{i=0}^{t-1} a_i(x).$$

For a fixed horizon m , define the 3-adic barrier-survivor set

$$\mathcal{S}_m = \{x \in U^* : A_t(x) < \alpha t \text{ for every } 1 \leq t \leq m\}. \quad (56)$$

Let

$$S_m = \mathcal{S}_m \cap \mathcal{S}$$

for ordinary positive admissible integers, and define the actual no-first-descent set

$$F_m = \{n \in \mathcal{S} : T^t(n) \geq n \text{ for every } 1 \leq t \leq m\}. \quad (57)$$

By the sufficient barrier theorem,

$$F_m \subseteq S_m. \quad (58)$$

10.2 Full cylinders and clopen representatives

For a finite symbol word

$$w = ((r_0, k_0), \dots, (r_{m-1}, k_{m-1})),$$

define the full cylinder

$$\tilde{C}_w = \phi_{r_0, k_0} \circ \dots \circ \phi_{r_{m-1}, k_{m-1}}(U), \quad (59)$$

and the symbol cylinder on the invariant domain

$$C_w = \tilde{C}_w \cap U^*.$$

Each $\tilde{C}_w$ is a residue class modulo a finite power of 3, hence is clopen in U .

Theorem 10.1 (Finite clopen representative). *For every fixed m , there exists a finite union of full cylinders*

$$\tilde{\mathcal{S}}_m \subset U$$

such that

$$\mathcal{S}_m = \tilde{\mathcal{S}}_m \cap U^*, \quad \mu(\mathcal{S}_m) = \mu(\tilde{\mathcal{S}}_m). \quad (60)$$

Moreover,

$$\tilde{\mathcal{S}}_m \cap \mathcal{S} = \mathcal{S}_m \cap \mathcal{S}. \quad (61)$$

Proof. The condition $A_m < \alpha m$ implies

$$A_m \leq \lceil \alpha m \rceil - 1.$$

Here αm is never an integer for $m \geq 1$. Indeed, if $\alpha m = N \in \mathbb{Z}$, then $4^m = 3^N$, which is impossible by unique prime factorization. Thus the strict real inequality has the exact integer cutoff displayed above. Since every valuation $a_i \geq 1$, only finitely many valuation words can satisfy all prefix inequalities through depth m , and for each valuation word there are only 2^m residue words. Their full cylinders form a finite clopen union $\tilde{\mathcal{S}}_m$. On U^* it represents exactly the survivor condition. Removing Z does not change Haar measure because Z is null. Finally, no positive integer belongs to Z , so the two sets agree on $\mathcal{S}$. $\square$

Theorem 10.2 (Clopen Haar measure equals relative natural density). *If $C \subseteq U$ is clopen, then*

$$\boxed{d_{\mathcal{S}}(C \cap \mathcal{S}) = \mu(C)}. \quad (62)$$

Proof. A clopen subset of U is a finite union of unit residue classes modulo 3^K for some K . There are

$$\varphi(3^K) = 2 \cdot 3^{K-1}$$

unit classes modulo 3^K . Each has normalized Haar mass

$$\frac{1}{2 \cdot 3^{K-1}},$$

and by elementary counting the same relative natural density among $\mathcal{S}$. Finite additivity gives the result. $\square$

Example 10.3 (A residue-class density modulo 9). Among the six unit residue classes modulo 9,

$$1, 2, 4, 5, 7, 8 \pmod{9},$$

each occurs with relative natural density $1/6$ inside $\mathcal{S}$. The same number is its normalized Haar mass in U . Finite unions of such classes therefore have identical Haar mass and relative natural density. The general modulus 3^K argument in the theorem is exactly the same counting principle.

Consequently,

$$d_{\mathcal{S}}(S_m) = \mu(S_m). \quad (63)$$

10.3 Finite discrepancy between barrier survival and actual no descent

The remaining issue is that S_m is defined by the sufficient barrier, whereas F_m records actual descent. The below-barrier correction term C_t could make a trajectory descend without crossing the barrier. At fixed depth, however, this difference affects only finitely many starting integers.

Theorem 10.4 (Finite discrepancy theorem). *For every fixed $m \geq 1$,*

$$\boxed{S_m \setminus F_m \text{ is finite.}} \quad (64)$$

Since $F_m \subseteq S_m$, equivalently

$$S_m \triangle F_m$$

is finite.

Proof. At fixed depth m , only finitely many barrier-surviving full symbol words occur. Fix such a word w . For each $1 \leq t \leq m$, the barrier inequality gives

$$3^{A_t(w)} < 4^t.$$

By the exact below-barrier criterion,

$$T^t(n) < n \iff n < R_t(w),$$

where

$$R_t(w) = \frac{C_t(w)}{4^t - 3^{A_t(w)}}.$$

For the fixed word define

$$R(w) = \max_{1 \leq t \leq m} R_t(w).$$

There are finitely many survivor words, so

$$M_m = \max_w R(w) < \infty.$$

Any $n \in S_m$ with $n \geq M_m$ cannot have an actual descent through time m . Hence every element of $S_m \setminus F_m$ lies below the finite bound M_m . $\square$

11 Exact finite-horizon first-descent density

Theorem 11.1 (Exact finite-horizon density theorem). *For every fixed $m \geq 1$,*

$$\boxed{d_{\mathcal{S}}(F_m) = d_{\mathcal{S}}(S_m) = \mu(\mathcal{S}_m).} \quad (65)$$

If

$$D_m = \{n \in \mathcal{S} : \exists 1 \leq t \leq m, T^t(n) < n\}, \quad (66)$$

then

$$\boxed{d_{\mathcal{S}}(D_m) = 1 - \mu(\mathcal{S}_m).} \quad (67)$$

Proof. By (63), S_m has relative density $\mu(\mathcal{S}_m)$. By the finite discrepancy theorem, S_m and F_m differ by a finite set, which has relative density zero. Therefore they have the same relative density. Since $D_m = \mathcal{S} \setminus F_m$, the second identity follows. $\square$

This theorem is the precise bridge between the 3-adic symbolic law and ordinary positive integers. The dynamic program below therefore computes an *exact asymptotic density*, not a sampling estimate.

12 Exact-rational dynamic programming

Let

$$p_t(s) = \mathbb{P}(A_t = s, A_u < \alpha u \text{ for all } 1 \leq u \leq t). \quad (68)$$

Initialize

$$p_0(0) = 1.$$

Because the valuations are iid with probability $2/3^k$, the recurrence is

$$p_{t+1}(s+k) += p_t(s) \frac{2}{3^k} \quad (69)$$

for every $k \geq 1$ satisfying

$$s + k < \alpha(t + 1). \quad (70)$$

Since $\alpha = \log_3 4$ is irrational, equality with an integer never occurs, and (70) is exactly equivalent to

$$s + k \leq \lceil \alpha(t + 1) \rceil - 1. \quad (71)$$

Finally,

$$\boxed{\mu(\mathcal{S}_t) = \sum_s p_t(s).} \quad (72)$$

Every transition probability is rational. The certification computation used exact rational arithmetic for all state masses through depth 500; floating-point arithmetic was not used to accumulate probabilities.

12.1 Certified finite-depth values

Table 1: Certified finite-depth barrier-survival probabilities and the corresponding exact finite-horizon first-descent densities. Exact fractions for the larger depths are displayed separately below to keep the table readable.

m	$\mu(\mathcal{S}_m)$	$d_{\mathcal{S}}(D_m) = 1 - \mu(\mathcal{S}_m)$
6	0.175582990397805213	0.824417009602194787
10	0.092488159551107272	0.907511840448892728
20	0.027761091757698629	0.972238908242301371
25	0.017778910915217290	0.982221089084782710
30	0.011001198894873564	0.988998801105126436
40	0.004976589672990874	0.995023410327009126
50	0.002371926204921184	0.997628073795078816

The exact rational values underlying the first seven rows are

$$\begin{aligned} \mu(\mathcal{S}_6) &= \frac{128}{729}, \\ \mu(\mathcal{S}_{10}) &= \frac{16384}{177147}, \\ \mu(\mathcal{S}_{20}) &= \frac{23521656832}{847288609443}, \\ \mu(\mathcal{S}_{25}) &= \frac{1220173365248}{68630377364883}, \\ \mu(\mathcal{S}_{30}) &= \frac{4953662807867392}{450283905890997363}, \\ \mu(\mathcal{S}_{40}) &= \frac{3572683711808203128832}{717897987691852588770249}, \\ \mu(\mathcal{S}_{50}) &= \frac{301646097509840481608007680}{127173474825648610542883299603}. \end{aligned}$$

At depth 50,

$$\boxed{\mu(\mathcal{S}_{50}) = \frac{301646097509840481608007680}{127173474825648610542883299603}.} \quad (73)$$

Numerically,

$$\mu(\mathcal{S}_{50}) \approx 0.00237192620492118432028189748,$$

and therefore

$$\boxed{d_{\mathcal{S}}(D_{50}) \approx 0.99762807379507881567971810252.} \quad (74)$$

Thus approximately

$$\boxed{99.76280737950788\%}$$

of admissible positive integers, in the sense of relative natural density, achieve a strict first descent within 50 reduced steps.

The same exact recurrence was continued to larger finite depths:

$$\mu(\mathcal{S}_{100}) \approx 8.5704175568195754 \times 10^{-5},$$

$$\mu(\mathcal{S}_{200}) \approx 2.6209977651649500 \times 10^{-7},$$

and

$$\mu(\mathcal{S}_{500}) \approx 3.0437604792015829 \times 10^{-14}.$$

These are exact-rational finite computations reported in decimal form. They do not, by themselves, prove any specific asymptotic equivalent.

13 Chernoff bound and the density-one first-descent theorem

For $t > 0$,

$$\mathbb{E}(e^{-ta}) = \sum_{k \geq 1} \frac{2e^{-tk}}{3^k} = \frac{2e^{-t}}{3 - e^{-t}}.$$

Put $q = e^{-t} \in (0, 1)$. Then

$$\mathbb{E}(q^a) = \frac{2q}{3 - q}.$$

Since $\mathcal{S}_m \subseteq \{A_m < \alpha m\}$,

$$\mu(\mathcal{S}_m) \leq \mathbb{P}(A_m < \alpha m).$$

For $q \in (0, 1)$, the function q^x decreases with x , so on the event $A_m < \alpha m$,

$$q^{A_m} > q^{\alpha m}.$$

Markov's inequality therefore gives

$$\mathbb{P}(A_m < \alpha m) \leq q^{-\alpha m} \mathbb{E}(q^{A_m}).$$

By independence,

$$\mathbb{E}(q^{A_m}) = (\mathbb{E}(q^a))^m = \left(\frac{2q}{3 - q} \right)^m.$$

Hence

$$\boxed{\mu(\mathcal{S}_m) \leq \left[\frac{2q^{1-\alpha}}{3 - q} \right]^m.} \tag{75}$$

Differentiating the logarithm of the bracket gives the minimizing value

$$\boxed{q^* = \frac{3(\alpha - 1)}{\alpha} \approx 0.622556248918265728.} \tag{76}$$

Define

$$\boxed{\rho = \frac{2(q^*)^{1-\alpha}}{3 - q^*} \approx 0.952392652605812657 < 1.} \tag{77}$$

Then

$$\boxed{\mu(\mathcal{S}_m) \leq \rho^m.} \tag{78}$$

Combining with the exact finite-horizon density theorem,

$$\boxed{d_{\mathcal{S}}(F_m) \leq \rho^m, \quad d_{\mathcal{S}}(D_m) \geq 1 - \rho^m.} \quad (79)$$

Define the eternal no-first-descent set

$$F_{\infty} = \bigcap_{m \geq 1} F_m. \quad (80)$$

Theorem 13.1 (Density-one first descent).

$$\boxed{d_{\mathcal{S}}(F_{\infty}) = 0.} \quad (81)$$

Equivalently,

$$\boxed{d_{\mathcal{S}}\{n \in \mathcal{S} : \exists j \geq 1, T^j(n) < n\} = 1.} \quad (82)$$

Proof. For every m ,

$$F_{\infty} \subseteq F_m.$$

Hence the upper relative density satisfies

$$\overline{d}_{\mathcal{S}}(F_{\infty}) \leq d_{\mathcal{S}}(F_m) \leq \rho^m.$$

Since $0 < \rho < 1$, letting $m \rightarrow \infty$ gives

$$\overline{d}_{\mathcal{S}}(F_{\infty}) = 0.$$

The lower density is nonnegative, so the relative natural density exists and equals zero. Taking the complement in $\mathcal{S}$ proves (82). $\square$

Remark 13.2 (Critical logical distinction). The theorem says that the set of admissible integers that never descend below their starting value has relative density zero. It does *not* say that this set is empty. Therefore it does not prove universal first descent. It also does not prove density-one convergence to the known terminal cycles: a first descent is only one local event in an orbit.

14 Exhaustive computation through 10^{11}

The analytic results above are complemented by a finite exhaustive computation. This section states exactly what was computed and exactly what was later revalidated.

14.1 Finite domain

The complete archived domain is

$$1 \leq n \leq 10^{11}, \quad 3 \nmid n.$$

Its exact cardinality is

$$10^{11} - \left\lfloor \frac{10^{11}}{3} \right\rfloor = 66,666,666,667. \quad (83)$$

The archive is the union of two disjoint computations:

$$1 \leq n \leq 10^9$$

and

$$10^9 < n \leq 10^{11}.$$

The extension contains exactly 66,000,000,000 admissible starts.

14.2 Known positive terminal cycles

The classification uses the three known positive terminal cycles

$$\{1\}, \quad \{2\}, \quad 22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22.$$

The term “known terminal cycles” is used deliberately. The finite computation does not prove that no additional positive cycle exists above the verified range.

14.3 Extension results

For $10^9 < n \leq 10^{11}$ the archived totals are

$$\text{tested} = 66,000,000,000,$$

$$\text{to } \{1\} = 6,536,905,587,$$

$$\text{to } \{2\} = 42,779,024,148,$$

$$\text{to four-cycle} = 16,684,070,265,$$

with

$$\text{unknown cycles} = 0, \quad \text{overflows} = 0.$$

The exact count identity is

$$6,536,905,587 + 42,779,024,148 + 16,684,070,265 = 66,000,000,000.$$

The extension also records

$$5,533,148,453,392$$

reduced-map iterations and total accumulated valuation

$$8,362,466,366,261.$$

Its record reduced transient is

$$574 \quad \text{at} \quad n = 97,312,923,676,$$

and its record accumulated valuation is

$$744 \quad \text{at the same start.}$$

The largest fully reduced state recorded in the extension is

$$580,315,803,007,928,858,285,$$

on the trajectory beginning at

$$n = 52,991,345,843.$$

Table 2: Complete archived classification for all $1 \leq n \leq 10^{11}$ with $3 \nmid n$.

Terminal cycle	Number of starts	Percentage
$\{1\}$	6,602,925,203	9.904388%
$\{2\}$	43,211,124,603	64.816687%
$22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22$	16,852,616,861	25.278925%
Total	66,666,666,667	100.000000%

14.4 Combined classification

Combining the base and extension archives gives Table 2.

The arithmetic check

$$6,602,925,203 + 43,211,124,603 + 16,852,616,861 = 66,666,666,667$$

is exact. Across the two archived component scans, no unknown positive cycle and no arithmetic overflow was reported.

Proposition 14.1 (Finite computational statement). *Within the archived finite domain*

$$1 \leq n \leq 10^{11}, \quad 3 \nmid n,$$

every tested start was classified into one of the three known positive terminal cycles, and the archived scans report zero unknown cycles and zero arithmetic overflows.

This is a finite computational statement, not an all-integers theorem.

15 Exhaustive first-descent verification through 10^{11}

Define the first-descent time, when it exists, by

$$\tau(n) = \min\{j \geq 1 : T^j(n) < n\}. \quad (84)$$

A dedicated first-lower scan was applied to the archived domain. For the extension $10^9 < n \leq 10^{11}$,

$$\text{first_lower_found} = 66,000,000,000, \quad \text{no_lower} = 0.$$

In the full range through 10^{11} , the scanner-semantic no-lower cases are exactly

$$1, 2, 8, 10, 13, 17, 22, 28, 29, 37, 38, 50. \quad (85)$$

These are small starts that enter or lie on a known terminal cycle before ever becoming strictly smaller than their own starting value. They are not counterexamples to convergence.

Proposition 15.1 (Finite first-descent verification). *For every positive integer n satisfying*

$$3 \nmid n, \quad 50 < n \leq 10^{11}, \quad (86)$$

the exhaustive computation finds a $j \geq 1$ such that

$$\boxed{T^j(n) < n.} \quad (87)$$

The largest observed first-descent time is

$$\boxed{\tau(76,089,719,024) = 355.} \quad (88)$$

16 The 355-step record and the descent barrier

The record orbit is particularly informative because its final first descent coincides with a crossing of the sufficient valuation barrier.

Across the first 355 transitions of the orbit beginning at

$$n = 76,089,719,024,$$

the valuation frequencies are

a	1	2	3	4	5
$\#$	283	54	14	3	1.

Therefore

$$A_{355} = 283 + 2(54) + 3(14) + 4(3) + 5 = 450.$$

Before the final step,

$$A_{354} = 445.$$

Numerically,

$$354\alpha \approx 446.6982655, \quad 355\alpha \approx 447.9601250. \quad (89)$$

Thus

$$A_{354} = 445 < 354\alpha,$$

whereas

$$A_{355} = 450 > 355\alpha.$$

The sufficient barrier is therefore not reached at step 354 but is reached at step 355.

The final five transitions are

$$1,399,884,236,737 \xrightarrow{a=2} 622,170,771,883,$$

$$622,170,771,883 \xrightarrow{a=2} 276,520,343,059,$$

$$276,520,343,059 \xrightarrow{a=1} 368,693,790,745,$$

$$368,693,790,745 \xrightarrow{a=1} 491,591,720,993,$$

and

$$491,591,720,993 \xrightarrow{a=5} 8,092,044,790.$$

For the final transition,

$$491,591,720,993 \equiv 2 \pmod{3},$$

so the correct residue is

$$r = 2.$$

Consequently,

$$4x - r = 4(491,591,720,993) - 2 = 1,966,366,883,970.$$

Exactly,

$$\nu_3(1,966,366,883,970) = 5,$$

and

$$\frac{1,966,366,883,970}{3^5} = 8,092,044,790.$$

Finally,

$$8,092,044,790 < 76,089,719,024,$$

which is the first strict descent below the initial value.

This example should not be interpreted as saying that every first descent occurs exactly when the sufficient barrier is crossed; the below-barrier criterion shows that descent can occur earlier. The record is useful because here the two events coincide.

17 Computational implementation and audit scope

Reproducibility requires distinguishing the historical exhaustive archive from later validation runs.

17.1 Scanner semantics

The audited scanner implements the exact map (4). A first-lower event is recorded only after a successful reduced-map step and only when the new state is strictly below the original start. Known terminal cycles are detected explicitly. A constant-memory cycle detector is used for a repeated state outside the known terminal cycles. Orbit arithmetic uses `unsigned __int128` with explicit finite-range overflow guards in the audited implementation.

17.2 V5.1 structural audit

The V5.1 source passed checks of:

- map arithmetic;
- first-lower semantics;
- known-terminal-cycle detection;
- unknown-cycle detection;
- multithreaded work allocation;
- audited finite-range overflow handling.

The audit nevertheless identified reproducibility weaknesses in checkpoint handling: a stale checkpoint could be accepted by filename without sufficiently strict validation of its stored interval identity; numeric parse success and required fields were not uniformly enforced; and a generic no-lower category could, in an abnormal termination path, mix resolved and unresolved cases.

17.3 V5.2 audit hardening

V5.2 was created to harden the audit layer without changing the dynamical rule. It adds:

- checkpoint format and version checks;
- validation of expected checkpoint index and interval endpoints;
- strict numeric parsing;
- required-field masks;
- aggregate consistency checks;
- separate known-no-lower and unresolved-no-lower categories;
- rejection of malformed or stale checkpoints.

17.4 Fresh validation runs

A fresh V5.2 run through 10^6 reproduced 666,667 admissible starts with

$$66,382 \rightarrow \{1\}, \quad 431,888 \rightarrow \{2\}, \quad 168,397 \rightarrow C_4,$$

and

$$\text{unknown} = 0, \quad \text{overflow} = 0.$$

It found

$$666,655$$

first-lower cases and the same 12 known no-lower exceptions.

Through 10^7 , V5.1 and V5.2 agreed on

$$6,666,667$$

admissible starts,

$$659,904 \rightarrow \{1\}, \quad 4,318,371 \rightarrow \{2\}, \quad 1,688,392 \rightarrow C_4,$$

with zero unknowns and zero overflows.

A fresh V5.2 re-execution of the production-scale raw interval

$$76,000,000,001 \leq n \leq 76,100,000,000$$

covered

$$66,666,667$$

admissible starts and reproduced the historical checkpoint statistics, including

$$n = 76,089,719,024, \quad \tau(n) = 355.$$

17.5 Checkpoint audit

The available archive contains 990 extension checkpoints and 10 base checkpoints. All

$$\boxed{1000/1000}$$

passed the applied structural checks, and independent aggregation recovered the published totals.

This statement is intentionally limited: it is a structural and aggregate validation of the available archive, not a cryptographic proof of the original historical bytes.

17.6 What was not rerun

The complete set of all

$$66,666,666,667$$

admissible starts through 10^{11} was *not* freshly recomputed from zero with V5.2 during the final audit. The full-domain statement relies on the exhaustive archived computation, its complete checkpoint collection, independent re-aggregation, structural auditing, and targeted fresh V5.2 re-executions.

18 How the analytic and computational results fit together

There are two independent notions of “large evidence” in this work.

First, the density theorem is asymptotic and analytic:

$$d_{\mathcal{S}}\{n : \exists j, T^j(n) < n\} = 1.$$

It says that the exceptional no-first-descent set has relative natural density zero, but it does not give a finite cutoff after which there are no exceptions.

Second, the exhaustive first-descent scan is finite and pointwise:

$$50 < n \leq 10^{11} \implies \exists j : T^j(n) < n.$$

It covers every admissible integer in a large finite interval, but says nothing directly about integers above 10^{11} .

Neither statement contains the other. Their agreement is nevertheless informative: the exact density theory predicts rapidly shrinking fixed-depth survivor density, while the finite scan finds no large exception through 10^{11} .

Similarly, the archived terminal-cycle classification through 10^{11} is much stronger than a sample, because every admissible start in that finite domain was included. But it is still finite and therefore cannot exclude a new cycle or an unbounded trajectory beginning above the verified range.

19 Open problems and conjectural directions

19.1 Universal first descent

The density-one theorem leaves open whether

$$F_{\infty} = \emptyset.$$

Equivalently, it remains open whether every admissible positive integer eventually falls below its starting value. The finite computation verifies this for every admissible n with $50 < n \leq 10^{11}$, but no proof is known for all n .

19.2 Global convergence

Conjecture 19.1 (Positive three-cycle convergence). Every $n \in \mathcal{S}$ eventually enters one of

$$\{1\}, \quad \{2\}, \quad 22 \rightarrow 29 \rightarrow 38 \rightarrow 50 \rightarrow 22.$$

The present results do not prove this conjecture. A complete proof must rule out both additional positive cycles and unbounded positive trajectories.

19.3 Density-one first descent versus density-one convergence

It is tempting to iterate the first-descent theorem recursively. Such an argument requires care. A density-zero exceptional set at one scale does not automatically remain negligible under the nonlinear preimage structure needed for repeated descent. No theorem in this paper establishes density-one convergence to the known terminal cycles.

19.4 Possible persistence asymptotics

The exact-rational values of $\mu(\mathcal{S}_m)$ decrease rapidly. Numerically they are compatible with a possible form

$$\mu(\mathcal{S}_m) \stackrel{?}{\sim} C m^{-3/2} \rho^m \quad (90)$$

for a constant $C > 0$, with ρ as in (77). No proof of (90) is given here. It should be regarded only as a numerical research direction.

19.5 Rigidity of low-valuation paths

A hypothetical no-descent orbit must remain permanently below the critical valuation line and in fact must satisfy the stronger deficit condition

$$\alpha j - A_j \rightarrow +\infty.$$

Understanding whether an ordinary integer orbit can realize such an infinite symbolic path is an integer-rigidity problem not resolved by Haar measure alone.

20 Claim boundaries and interpretation

For clarity, the status of the main statements is summarized in Table 3.

Table 3: Logical status of the principal claims.

Statement	Status
Exact inverse branches and Haar scaling	Theorem
Haar invariance	Theorem
$\mathbb{P}(a_i = k) = 2/3^k$ and iid valuations	Theorem on U^*
Exact multi-step identity	Deterministic theorem
$A_j \geq j \log_3 4 \Rightarrow T^j(n) < n$	Deterministic theorem
Exact below-barrier descent criterion	Deterministic theorem
Summability and deficit conditions under eternal no descent	Conditional deterministic theorems
Finite clopen representative and Haar-to-integer transfer	Theorem
$S_m \triangle F_m$ finite	Theorem
$d_{\mathcal{S}}(F_m) = \mu(\mathcal{S}_m)$	Theorem
Density-one first descent	Main theorem
DP values through depth 500	Exact-rational finite computation
99.7628073795% first descent within 50 steps	Certified consequence of theorem + exact DP
Classification through 10^{11}	Finite exhaustive computational statement
$\tau_{\max} = 355$ through 10^{11}	Finite computational record
Universal first descent	Open
Density-one convergence to known cycles	Not established
Universal convergence to the three known positive terminal cycles	Conjecture
Absence of unbounded positive trajectories	Open
Asymptotic (90)	Numerical/conjectural only

21 Conclusion

The map

$$T(x) = \frac{4x - r(x)}{3^{\nu_3(4x - r(x))}}, \quad r(x) = x \bmod 3 \in \{1, 2\},$$

combines a fixed multiplicative expansion by 4 with a variable exact 3-adic compression. The present analysis separates its behavior into an exact symbolic component, an exact deterministic orbit component, an asymptotic density component, and a finite computational component.

On the 3-adic unit space, the inverse branches are explicit and their Haar scaling is exact. This proves that the successive valuations are iid with

$$\mathbb{P}(a = k) = \frac{2}{3^k}, \quad \mathbb{E}[a] = \frac{3}{2}.$$

For ordinary positive integer orbits, the exact multi-step identity

$$T^j(n) = \frac{4^j n - C_j}{3^{A_j}}$$

produces the deterministic sufficient descent barrier

$$A_j \geq j \log_3 4.$$

If an orbit were never to fall below its start, it would have to satisfy the much stronger restrictions

$$\sum_j \frac{3^{A_j}}{4^j} < \infty, \quad j \log_3 4 - A_j \rightarrow +\infty,$$

together with the additional series, product, and congruence identities derived above.

At fixed depth, the 3-adic barrier-survivor condition is represented by finitely many clopen cylinders. Its Haar measure equals the relative natural density of the corresponding integer residue classes, and the difference between barrier survival and actual no descent is finite. This yields the exact identity

$$d_{\mathcal{S}}(F_m) = \mu(\mathcal{S}_m).$$

The exact-rational dynamic program then gives concrete certified densities, including approximately 99.7628073795% first descent within 50 reduced steps. The optimized Chernoff bound implies

$$d_{\mathcal{S}}(F_{\infty}) = 0,$$

which is the density-one first-descent theorem.

The finite computation provides a complementary pointwise result. The archived exhaustive domain through 10^{11} contains 66,666,666,667 admissible starts, all classified into the two known fixed points or the known four-cycle, with zero reported unknown cycles and zero overflows. Every admissible $50 < n \leq 10^{11}$ achieves a strict first descent; the largest observed first-descent time is 355 at 76,089,719,024. The final step of that record has residue $r = 2$, valuation 5, and lands at 8,092,044,790.

These results substantially narrow the form of any possible counterexample, but they do not eliminate all counterexamples. Universal first descent, density-one convergence, exclusion of all additional positive cycles, exclusion of unbounded trajectories, and the global three-terminal-cycle convergence conjecture remain open.

A Exact-rational DP pseudocode

A minimal exact implementation of the finite-horizon recurrence can be described as follows.

```

p = {0 : 1}                                # exact rational masses
for t = 1, 2, ..., m:
    cap = ceil(alpha*t) - 1
    new = empty map
    for each (s, mass) in p:
        for k = 1, ..., cap-s:
            new[s+k] += mass * (2 / 3^k)
    p = new
mu_S_t = sum of all masses in p

```

For certification, the masses $2/3^k$ and all accumulated probabilities are stored as exact rational numbers. The only nonrational quantity is the barrier slope $\alpha = \log_3 4$, and because αt is never an integer for positive integer t , the admissible integer threshold is unambiguously

$$\lceil \alpha t \rceil - 1.$$

The certification run generated these thresholds at high precision and cross-checked them independently through depth 500.

B Additional finite computational totals

For the base range $1 \leq n \leq 10^9$:

$$\begin{aligned}
 \text{admissible starts} &= 666,666,667, \\
 \text{to } \{1\} &= 66,019,616, \\
 \text{to } \{2\} &= 432,100,455, \\
 \text{to four-cycle} &= 168,546,596.
 \end{aligned}$$

The base record reduced transient is 405 at

$$n = 833,560,375,$$

and the base record accumulated valuation is 529 at the same start. The largest fully reduced base state is

$$164,210,561,845,359,809$$

from

$$n = 664,934,564.$$

For the extension $10^9 < n \leq 10^{11}$:

$$\text{total reduced steps} = 5,533,148,453,392,$$

$$\text{total accumulated valuation} = 8,362,466,366,261.$$

The combined basin counts are exactly the sums of the two disjoint ranges.

C Why the twelve small no-lower cases are not counterexamples

The first-descent scanner asks whether an orbit ever becomes *strictly smaller than its own starting value* before it terminates in a known cycle. This is different from asking whether the orbit converges.

For example, 1 and 2 are fixed points, so they never become strictly smaller than themselves. Likewise, a start lying on the four-cycle can circulate without visiting a value below its own start before the scanner recognizes the known terminal cycle. Other small starts in

$$1, 2, 8, 10, 13, 17, 22, 28, 29, 37, 38, 50$$

enter a known terminal cycle before a strict first-lower event relative to their starting value. Thus “no lower” in this scanner-specific sense is not synonymous with “nonconvergent.”

D Reproducibility notes

The final computational audit distinguishes three layers:

1. the original exhaustive archived scans;
2. independent structural checking and re-aggregation of the complete available checkpoint collection;
3. fresh execution of the audit-hardened V5.2 scanner on validation ranges and on a production-scale interval containing the hardest first-descent record.

The source-level hardening was designed to improve reproducibility and checkpoint integrity rather than to alter the mathematical map. Any future full rerun should preserve the exact map, terminal-cycle definitions, first-lower semantics, and integer arithmetic while retaining strict checkpoint identity and parsing checks.

E Selected numerical constants

For reference,

$$\begin{aligned}\alpha &= \log_3 4 \approx 1.26185950714291487419905422868552, \\ \beta &= \alpha - 1 \approx 0.26185950714291487419905422868552, \\ q^* &= \frac{3(\alpha - 1)}{\alpha} \approx 0.62255624891826572781939158407828,\end{aligned}$$

and

$$\rho \approx 0.95239265260581265703694506264349.$$

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